

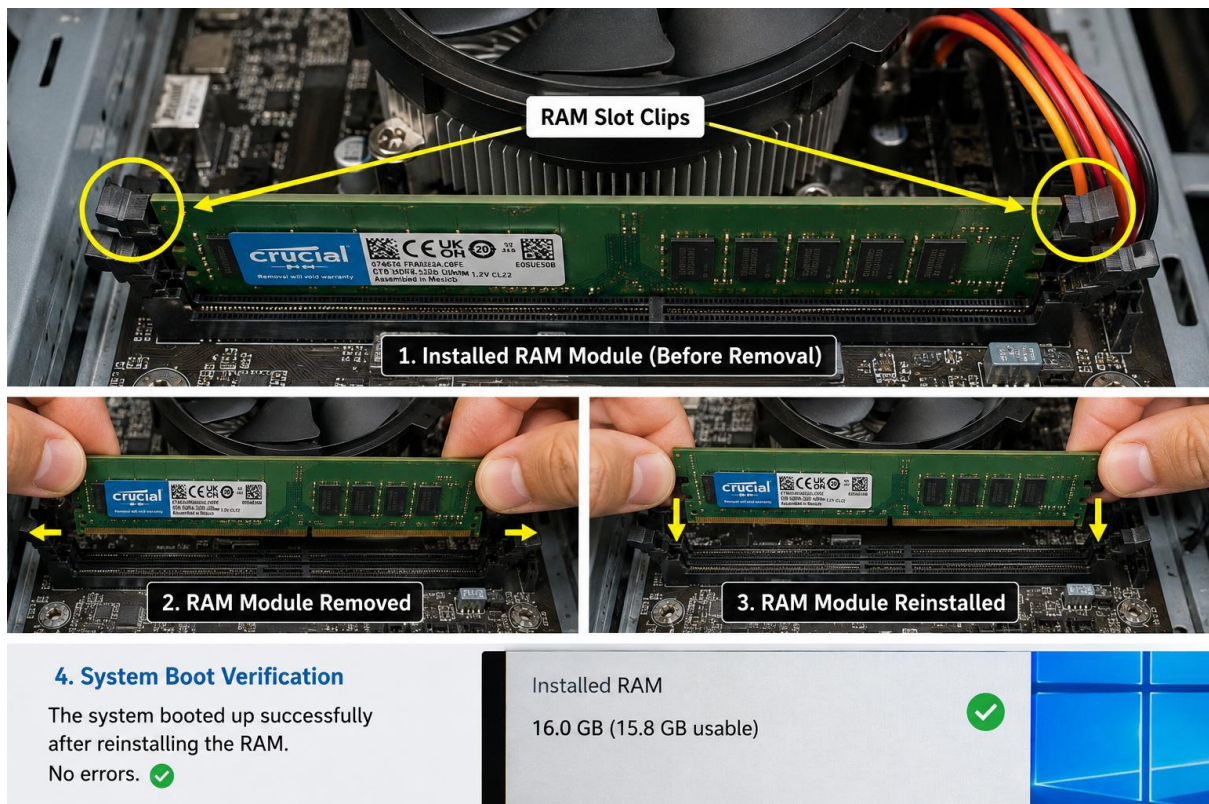
ASSIGNMENT-1.3

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Speed	800 – 2133 MT/s	1600 – 3200+MT/s	4800 – 8400+MT/s
Voltage	1.5v	1.2v	1.0v
Max capacity per stick	Up to 8GB – 16GB	Up to 32GB – 64GB	Up to 128GB – 256GB
Pin count	240 pins	288 pins	288 pins (different notch position)
Typical use cases	Older PCs, basic home/office tasks	Mainstream desktop PCs, office workstations	High-rang gaming PCs, heavy video editing, 3D rendering, modern servers

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD (Hard Disk Drive)	SATA SSD (Solid State Drive)	NVMe (Non-Volatile Memory Express)
Technology & Interface	Uses spinning magnetic platters and moving write heads.	Uses microchip-based flash memory connected via a SATA interface.	Uses flash memory connected directly to high-speed PCIe motherboard slots.
Speed	Slowest (80 MB/s to 160 MB/s).	Fast (500 MB/s to 550 MB/s).	Ultra-fast (3500 MB/s to 7000 MB/s).
Cost per GB	Cheapest per Gigabyte.	Moderate price point.	Highest cost per Gigabyte due to high performance.

- Three key differences

- Daily Apps & Use Cases

1. HDD (Hard Disk Drive)

- **App/Use Case:** long term storage (windows File Explore or Google Drive Sync for storing large photo albums, movies, and system backups).
- **Benefit:** Provides massive storage capacity at a low price where high speed isn't critical.

2. SATA SSD

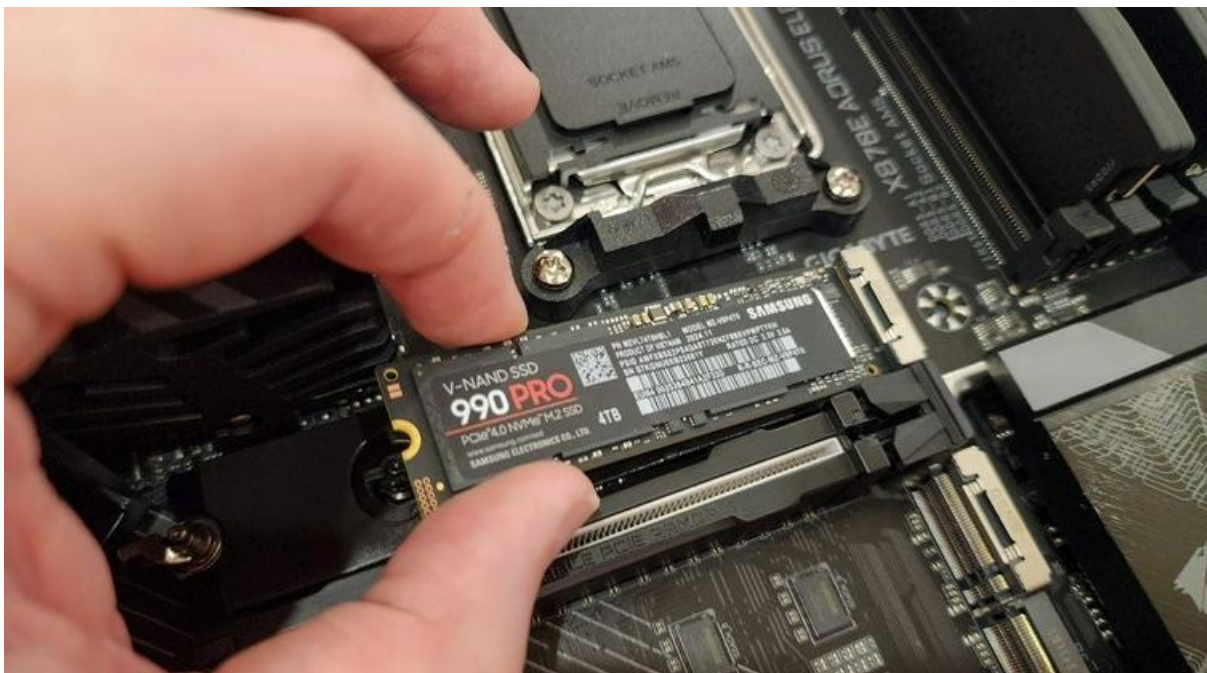
- **App/Use Case:** Everyday productivity & Web Browsing (Google Chrome, Microsoft Word / Excel).
- **Benefit:** Ensures fast operating system boot times and instant app launching for smooth daily use without high costs.

3. NVMe SSD

- **App/Uses Case: Heavy Video Editing & High-End Gaming**
- **Benefit: Seamlessly streams high-bitrate 4K/8K video timelines and transfers multi-gigabyte project files instantly without lag or loading stutters.**

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.



Objective:

- To remove the storage drive, identify its connector type, reinstall it correctly, and verify that the system detects the drive after reassembly.

Hardware Specifications:

- Motherboard: ASUS ROG Maximus Z790 Formula
- Storage Drive: Samsung 990 Pro 2TB M.2 NVMe SSD

Step-by-Step Procedure:

✓ Drive Removal (Panel 1): The computer was powered off and disconnected from the power supply. The M.2 heatsink was removed carefully, and the SSD was taken out from the motherboard.

✓ Storage Drive Identification (Panel 2): The storage drive connector was identified as an M.2 NVMe SSD connector.

✓ Drive Reinstallation (Panel 3): The SSD was aligned with the M.2 slot and inserted correctly. The retaining screw and heatsink were reinstalled securely.

✓ System Verification (Panel 4): After reassembling the computer, it was powered on. Windows successfully detected the SSD, and the drive appeared in Disk Management and File Explorer.

Result: The storage drive was successfully reinstalled, and the system recognized it correctly without any errors.